Analytical Chemistry Cumulative Exam

January 27th, 2012

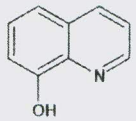
Open-Ended Analytical Chemistry Problem Solving

Directions: Pick **one** problem from each class of problems listed below.

Class 1. Equilibrium Chemistry.

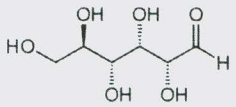
1. Explain how you would make a constant ionic strength acetic acid buffer (0.2 M ionic strength) with a pH of 5.0. For acetic acid, the pK_a is 4.75.

2. For 8-hydroxyquinolone (oxime), $K_{a1} = 8 \times 10^{-6}$ and $K_{a2} = 1.4 \times 10^{-10}$ (you should know which groups these are based on your cumulative knowledge of chemistry). The K_d for distribution into chloroform from water is 720. Calculate the distribution ratio (*D*) for oxime at pH 12.

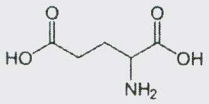


Class 2. Instrumentation.

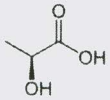
1. Fermentation plants commonly require continuous monitoring of inputs such as glucose and glutamic acid and outputs including lactic acid. Since fermentation involves cells and other proteins, the best way to measure this would be non-invasively using an optical method. Describe an optical method that could be used for such a process. Explain briefly the fundamental principles of such a method and the methods for calibration that would have to be employed.



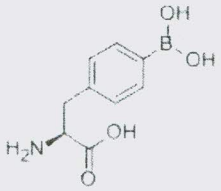
Glucose



Glutamic Acid



Lactic Acid

2. For the above-mentioned problem, describe how you would fully validate this method. Describe the basics of all the instrumentation to be used and the criteria for validation.

3. L-boronophenylalanine (shown below) is one of many compounds used for boron neutron capture therapy (BNCT) for different tumors. Assume you are on the analytical chemistry team for a company making a new boron-containing compound for BNCT. Determining the total distribution (concentration and tissue location) of the new drug to all tissues and fluids within experimental animals is a prerequisite for ultimately performing human studies. Describe how you would selectively measure L-boronophenylalanine for a biodistribution study. Be sure to mention sample preparation and the logic behind your choices of instrumentation. There are different methods for achieving the goal set forth in this problem. However, maximum credit will be given to answers that have identified the most selective instrumentation for this task and the chemical logic for choosing that instrumentation.